

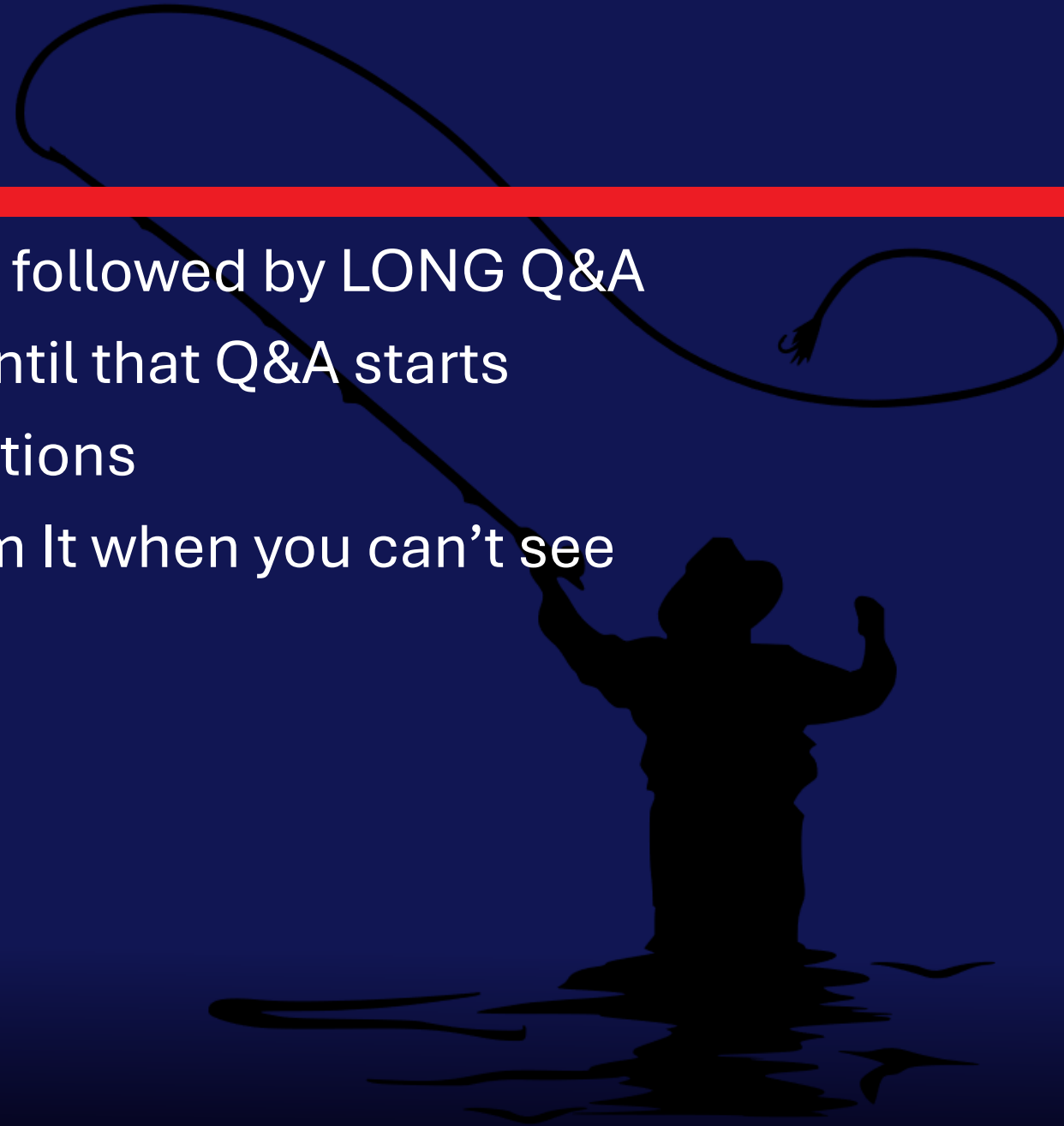
Build Your Test Lab:

No Lab? No Problem!



Attention

- ◆ MMS sessions are 60-75 minutes followed by LONG Q&A
- ◆ Please hold detailed questions until that Q&A starts
- ◆ Feel free to ask clarification questions
- ◆ Remind the speakers to use Zoom It when you can't see



Speakers



Mattias Benninge

Principal Engineer
2Pint Software

 @matbg



Mike Terrill

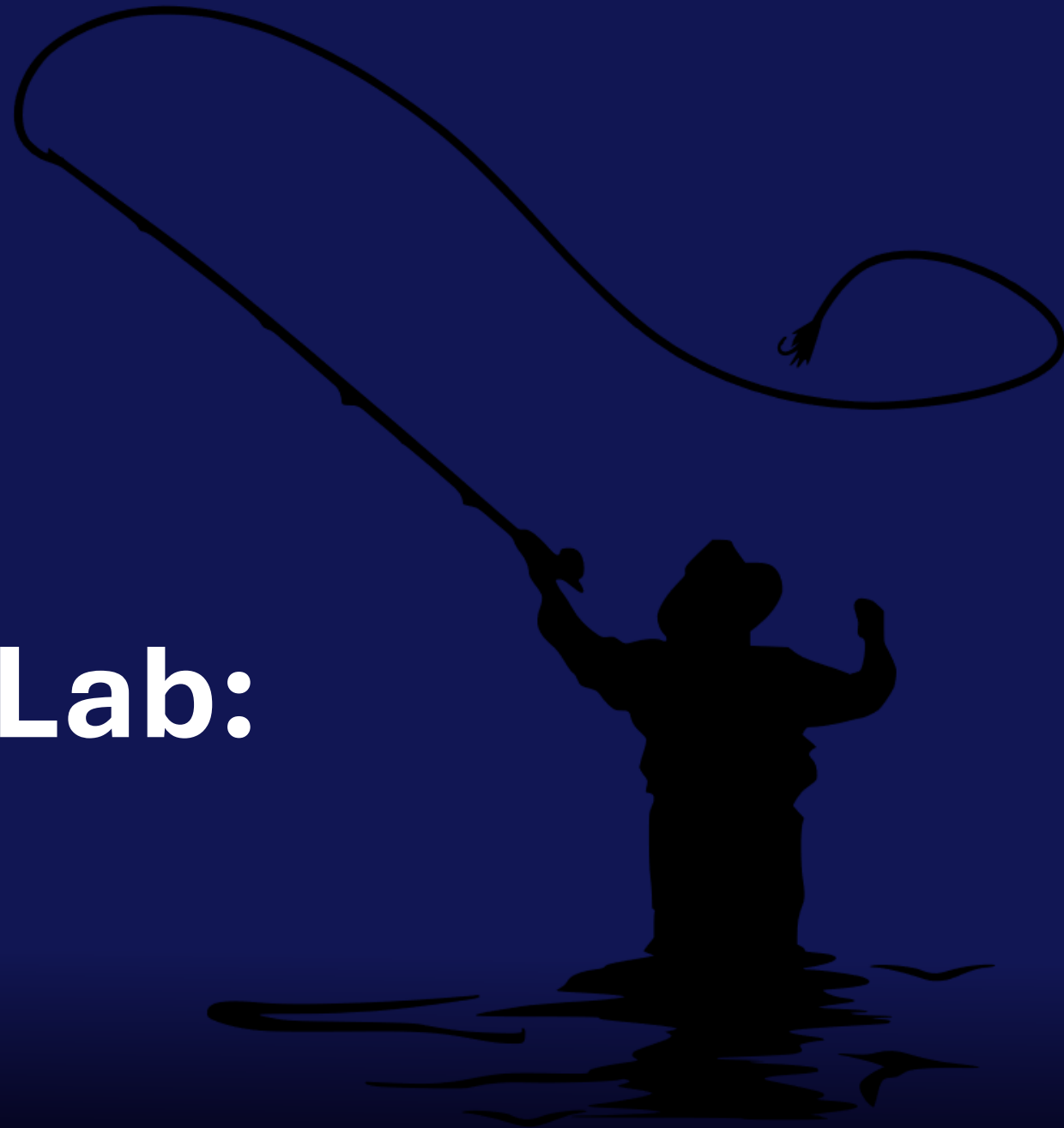
Principal Engineer, Microsoft MVP
2Pint Software

 @miketerrill

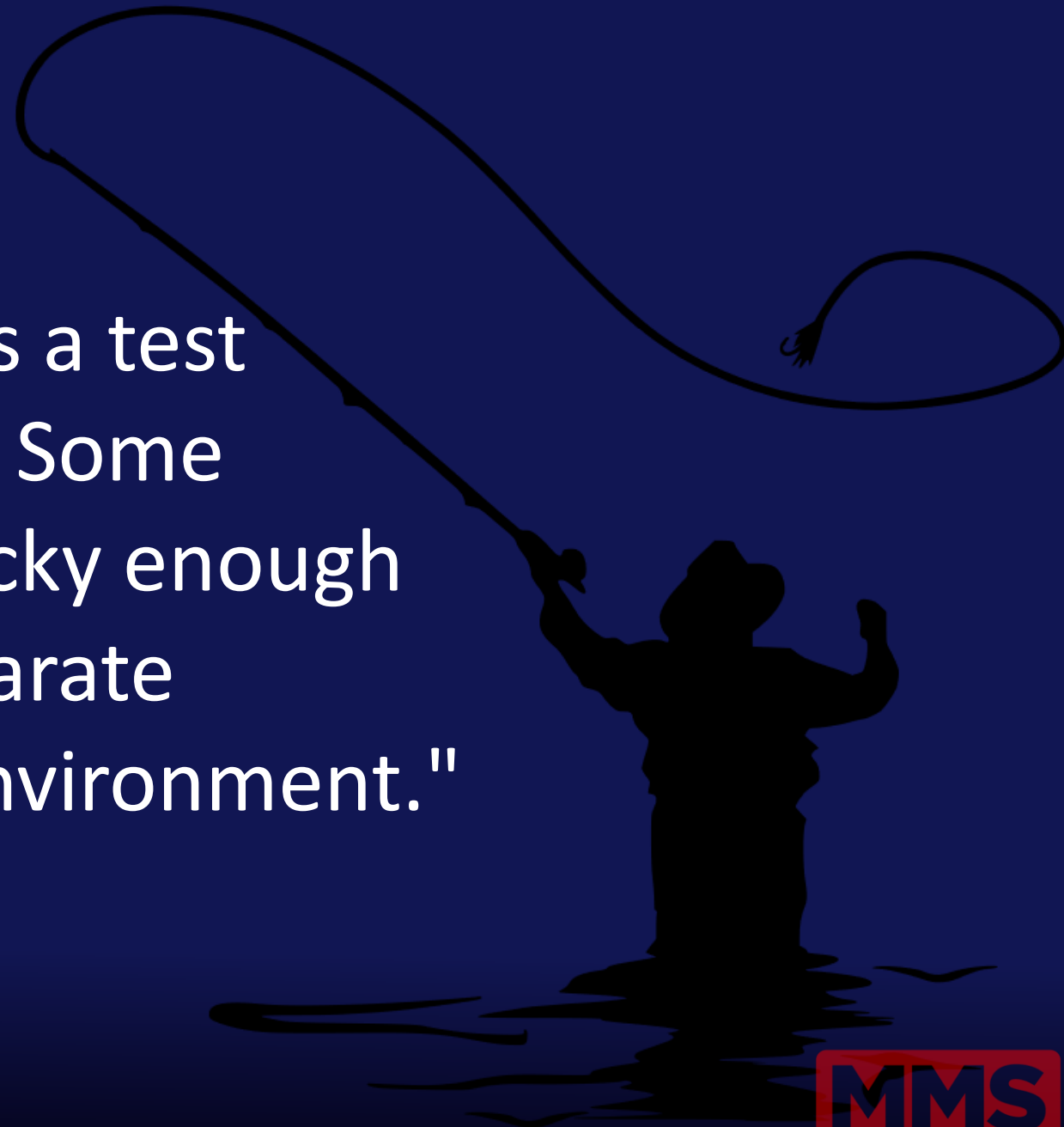
Can't see our slides? Can't hear? Need to repeat the question? Call us out!

Build Your Test Lab:

No Lab? No Problem!



"Everyone has a test environment. Some people are lucky enough to have a separate production environment."



Why You Should Have a Lab

- ◆ Never, ever test in production! (RGE)
- ◆ Work Labs
 - ◆ Should replicate production
 - ◆ Good hardware collection
 - ◆ Test here first
- ◆ Home Labs
 - ◆ Hone your skills, learn new technology
 - ◆ Virtual machines are perfect



Lab Software

- ♦ Hydration Kit by Johan
 - ♦ <https://www.deploymentresearch.com/hydration-kit-for-windows-server-2022-sql-server-2019-and-configmgr-current-branch/>
 - ♦ Advantages – Customize to your environment.
 - ♦ Disadvantages – Takes some time to setup.
- ♦ Windows and Office 365 Deployment Lab Kits from Microsoft
 - ♦ Windows 11: <https://www.microsoft.com/en-us/evalcenter/evaluate-windows-11-office-365-lab-kit>
 - ♦ Advantages – Setup quickly.
 - ♦ Disadvantages – Hardcoded to corp.Contoso.com
- ♦ Build Your Own
 - ♦ Configuration Manager OSD
 - ♦ DeployR/DeployR Community Edition



Low end (OSD) vs High End (OSD)

Single Virtual Machine

- ◆ Use your existing Infrastructure
- ◆ Create a Pre-Prod Task Sequence
- ◆ Test the Client on a Virtual Machine
- ◆ Virtual Machines are great for testing!

Full Lab Environment

- ◆ Full Virtualized Environment
 - ◆ Domain Controller
 - ◆ Gateway/Router
 - ◆ ConfigMgr Server
 - ◆ Distribution Point
 - ◆ Test Clients
- ◆ Augmented w/ Physical Clients

Hardware

◆ Processors:

- ◆ i7 or Newish i5
- ◆ i7 8th or better for Windows 11
- ◆ Used Xeon Scalable are good buy
- ◆ Ryzen/ThreadRipper
- ◆ 8 Cores – 16 threaded
- ◆ See: <https://cpubenchmark.net>

◆ RAM:

- ◆ 32GB (min) 64GB+ (Optimal)
- ◆ Servers: 64GB+

◆ Storage:

- ◆ 1tb NVME SSD - King for VM's
- ◆ Put your code data on HDD
- ◆ See: <https://shucks.top>
- ◆ See: <https://diskprices.com>

◆ Networking:

- ◆ 1Gbps (min)
- ◆ 10GB SFP+ Used ConnectX-3 w/ DAC - Good 4 testing
- ◆ 2.5gb is the new thing
<https://amazon.com/s?k=2.5gb+switch>

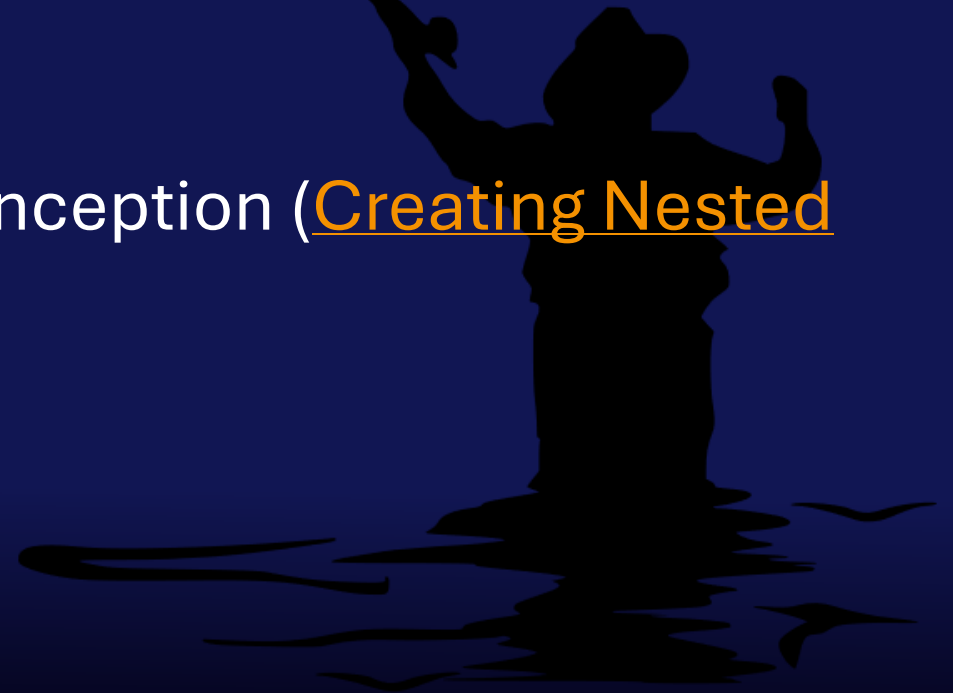
Host Examples

- ◆ Cheap – Upgrade
 - ◆ Re-Use an old Gaming PC, add an extra NVME drive.
 - ◆ Add some more RAM Existing Laptop or Desktop!
- ◆ Moderate - Used
 - ◆ Search on E-Bay – The [GaryTown](#) Method
 - ◆ Search for Workstations on E-Bay or <https://PCServerAndParts.com>
 - ◆ Get used PC's with CPU and RAM. Get your own disks.
- ◆ All New
 - ◆ Super Small - [Minisforum MS-01](#)
 - ◆ [Fanless](#) Firewall Appliances w/ Intel n100



Setting up a Virtual Machine

- ♦ 4 CPU's ([Installing updates during Autopilot: Windows 11 edition, revisited again – Out of Office Hours \(oofhours.com\)](#))
- ♦ Dynamic vs Static Memory
- ♦ Checkpoints
- ♦ Secure Boot
- ♦ Nested Virtual Machines – aka Hyper-V inception ([Creating Nested Hyper-V VMs - Deployment Research](#))



Software

◆ Network Software

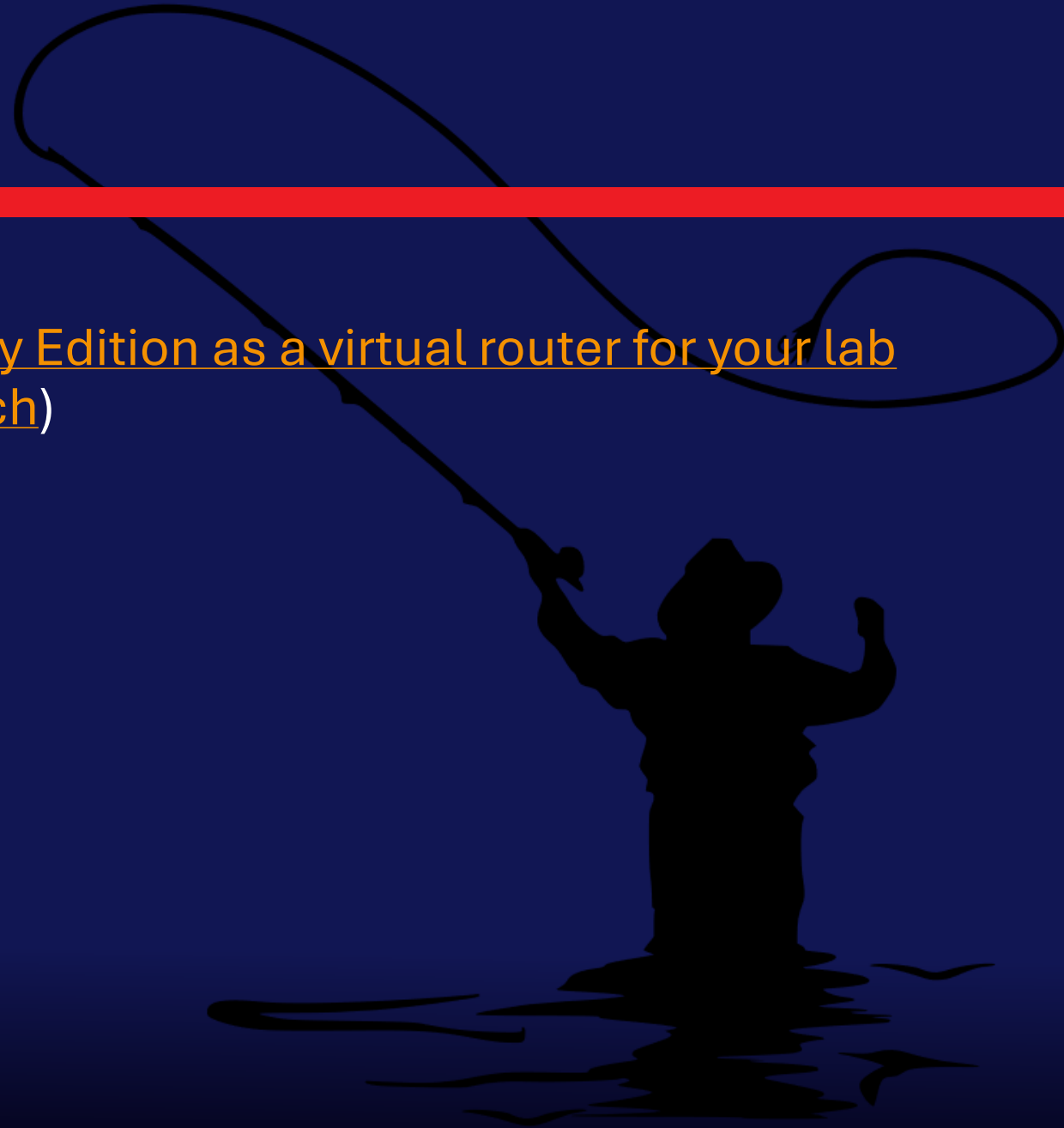
- ◆ pfSense ([Using pfSense Community Edition as a virtual router for your lab environment - Deployment Research](#))
- ◆ TailScale
- ◆ TinyPilot (KVM)

◆ HyperVisors

- ◆ Hyper-V
- ◆ VMWare
- ◆ ProxMox

◆ Storage

- ◆ Storage Spaces



Intune LAB Tenant

◆ 30 day trial (Well....)

- + Free
- Only last 30 days...

◆ Microsoft 365 Developer Program

- + 25 Free licenses
- Need a Visual Studio Pro (or better) Subscription
- Not renewed unless doing active development
- Pay as you go subscription not available (No MCC)

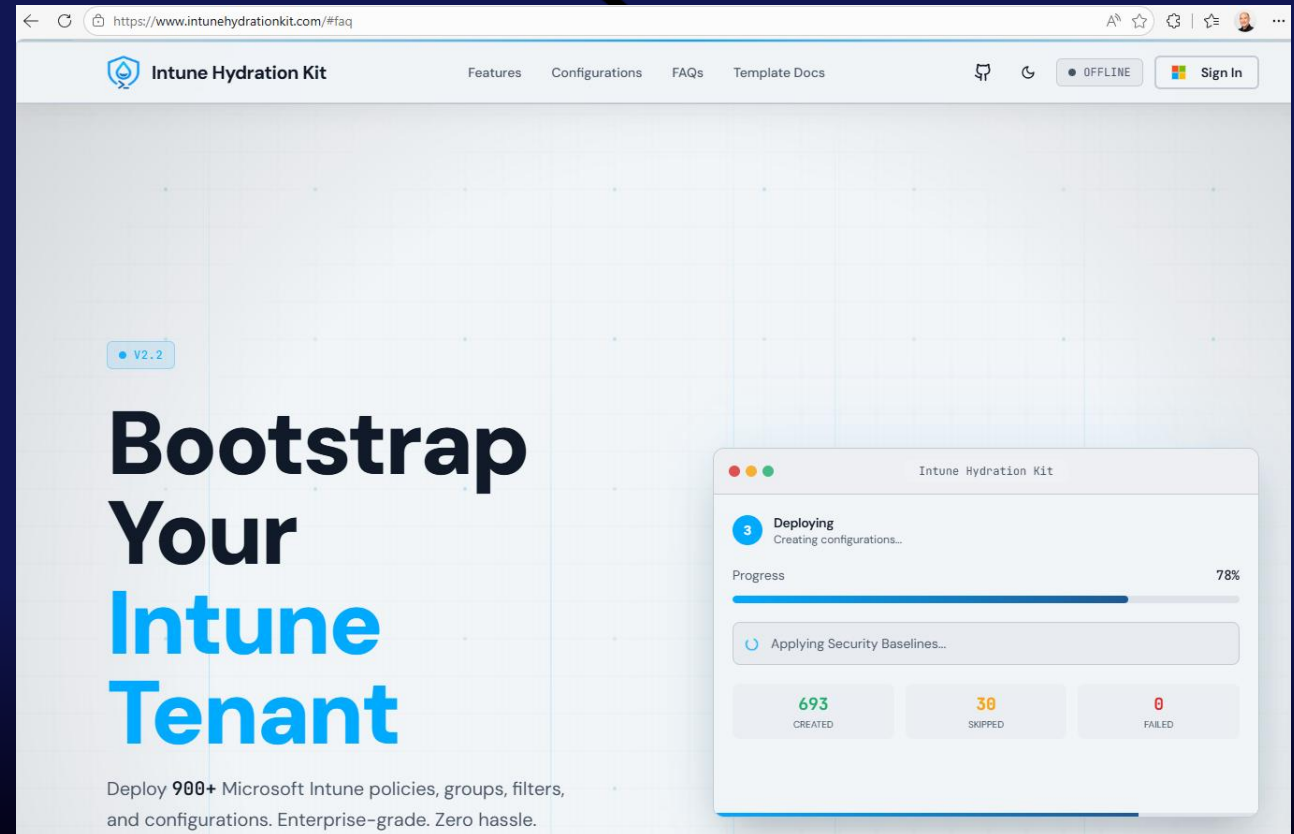
◆ Payed

- + Keep as long as you pay
- Cost



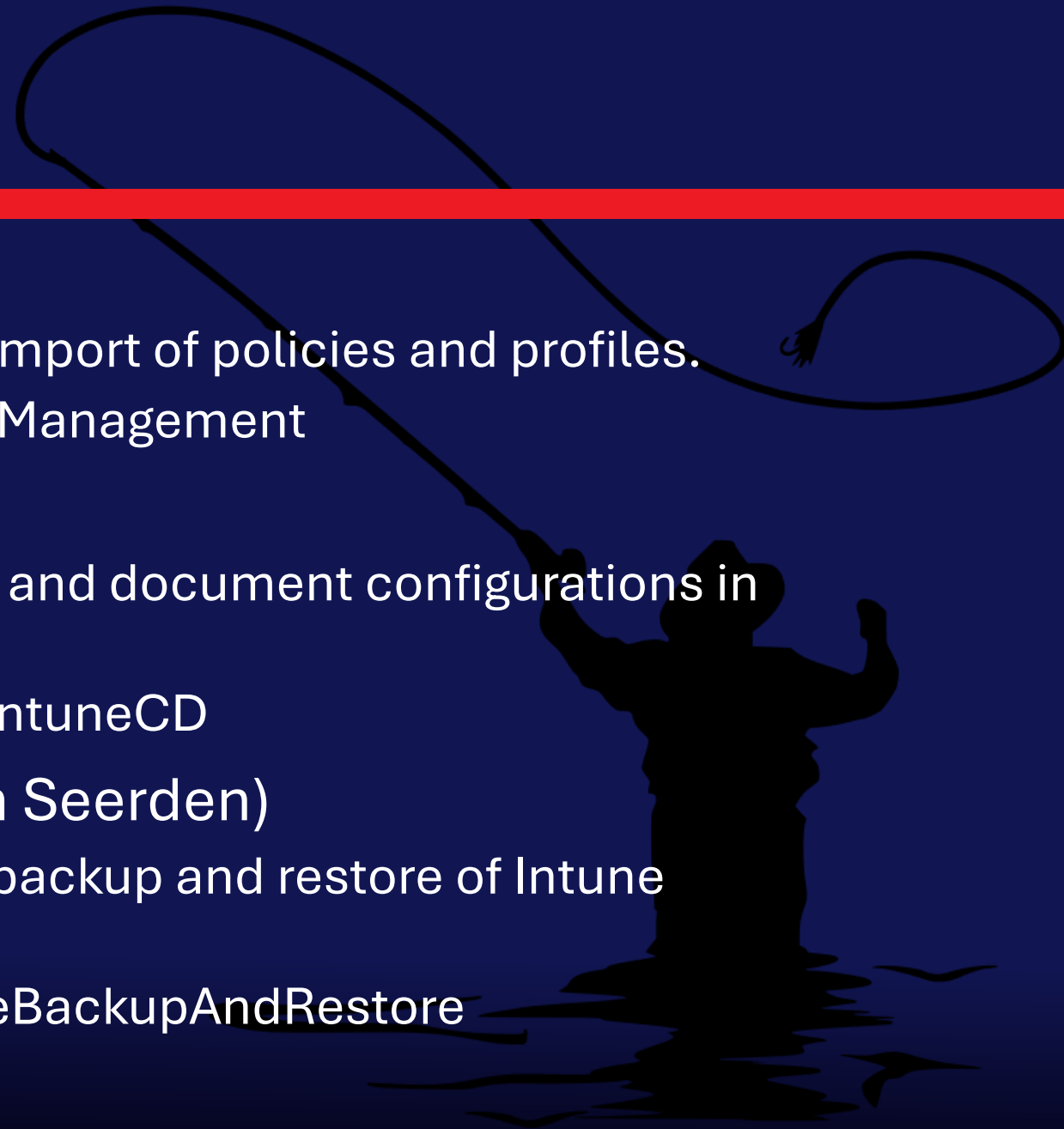
Intune Tools

- ♦ Intune Hydration Kit (by Jorge Suarez)
 - ♦ Good if starting with a fresh lab
 - ♦ <https://www.intunehydrationkit.com/>



Intune Tools

- ◆ Intune Management (by Micke K)
 - ◆ PowerShell tool to simplify Export/Import of policies and profiles.
 - ◆ <https://github.com/Micke-K/IntuneManagement>
- ◆ IntuneCI (by Tobias Almén)
 - ◆ Automation tool to backup, update and document configurations in Intune.
 - ◆ <https://github.com/almenscorner/IntuneCD>
- ◆ Intune Backup & Restore (by John Seerden)
 - ◆ Multitenant PowerShell module to backup and restore of Intune configurations.
 - ◆ <https://github.com/jseerden/IntuneBackupAndRestore>



DEMO

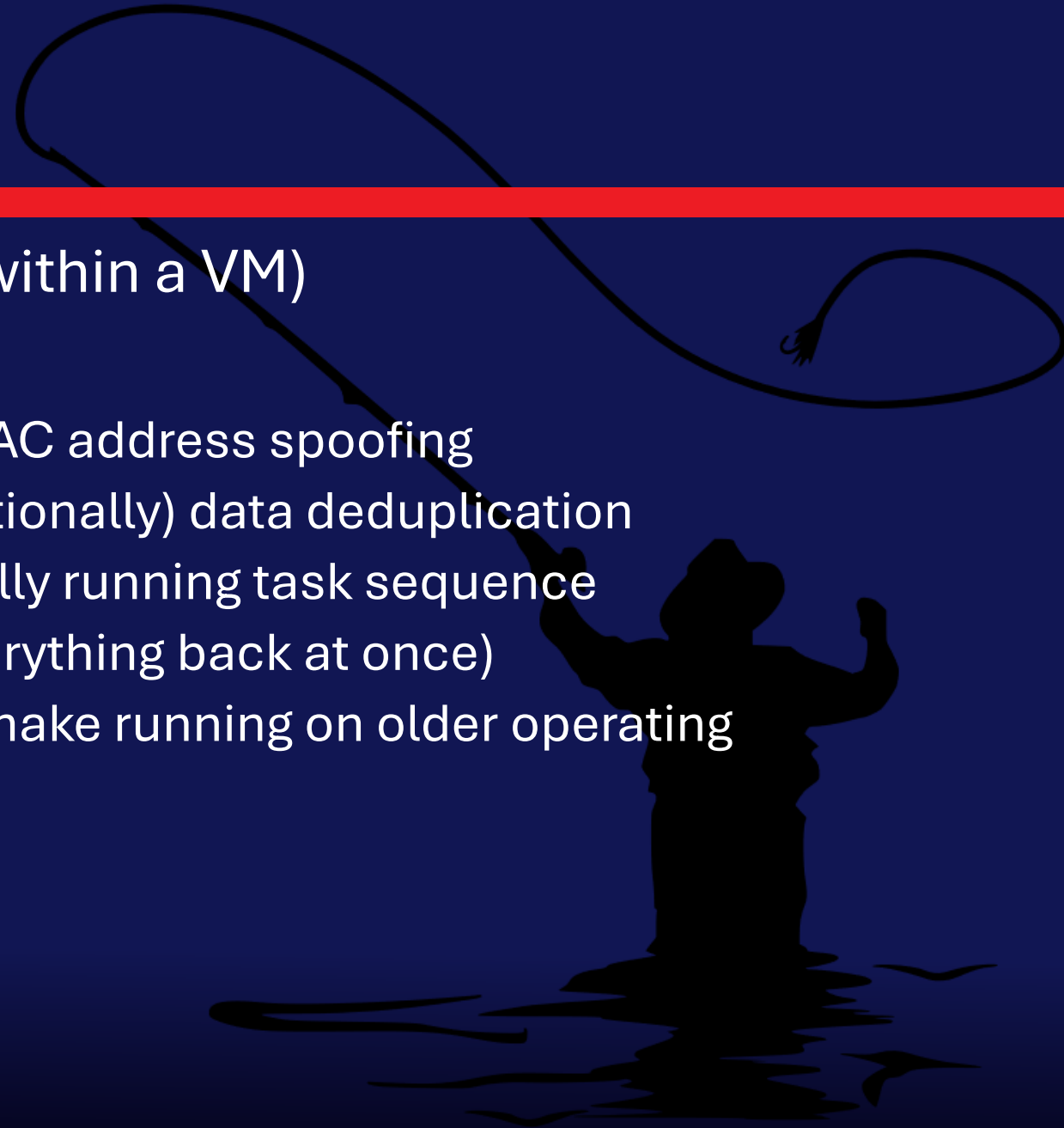
DEMO

Intune Hydration Kit



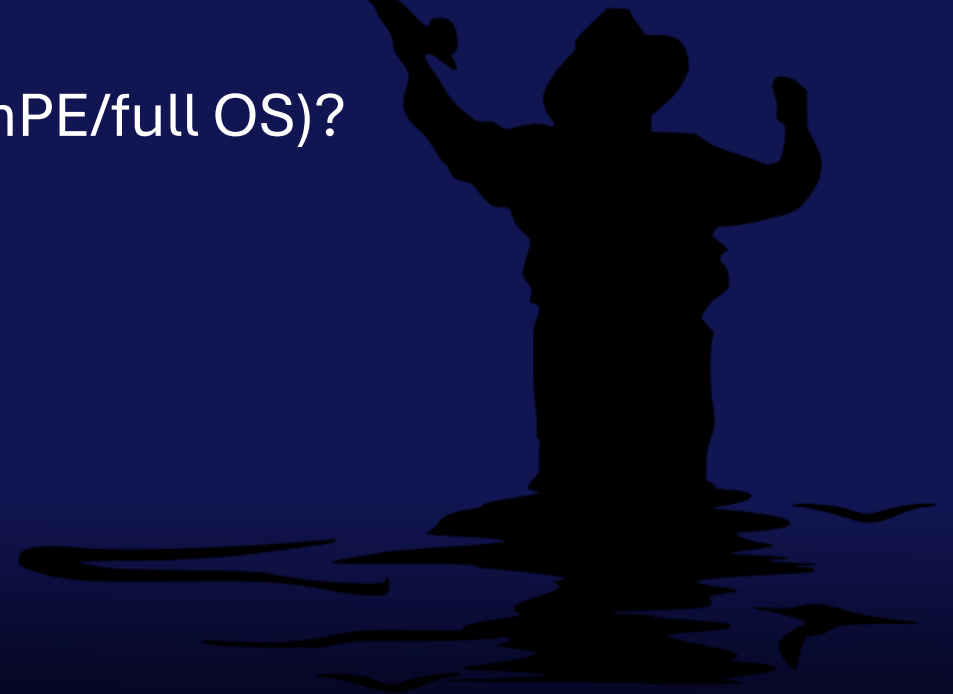
Build your own lab

- ◆ Hyper-V Inception (VMs running within a VM)
- ◆ Create parent VM:
 - ◆ Enable nested virtualization and MAC address spoofing
 - ◆ Use Server (Eval), 2 drives, and (optionally) data deduplication
 - ◆ Static MAC address for automatically running task sequence
 - ◆ Use standard check points (roll everything back at once)
 - ◆ Set configuration version lower to make running on older operating systems easier



Build your own lab (cont.)

- ◆ Parent VM:
 - ◆ Install and configure Hyper-V
 - ◆ Create child VMs (servers/clients/DHCP/DNS)
 - ◆ Static MAC address for automatically running task sequence
 - ◆ Boot and build child VMs
 - ◆ Isolated network - what about DHCP (in WinPE/full OS)?
- ◆ Child VMs:
 - ◆ Domain Controller/DHCP/DNS server
 - ◆ Other server(s)
 - ◆ Clients



DEMO

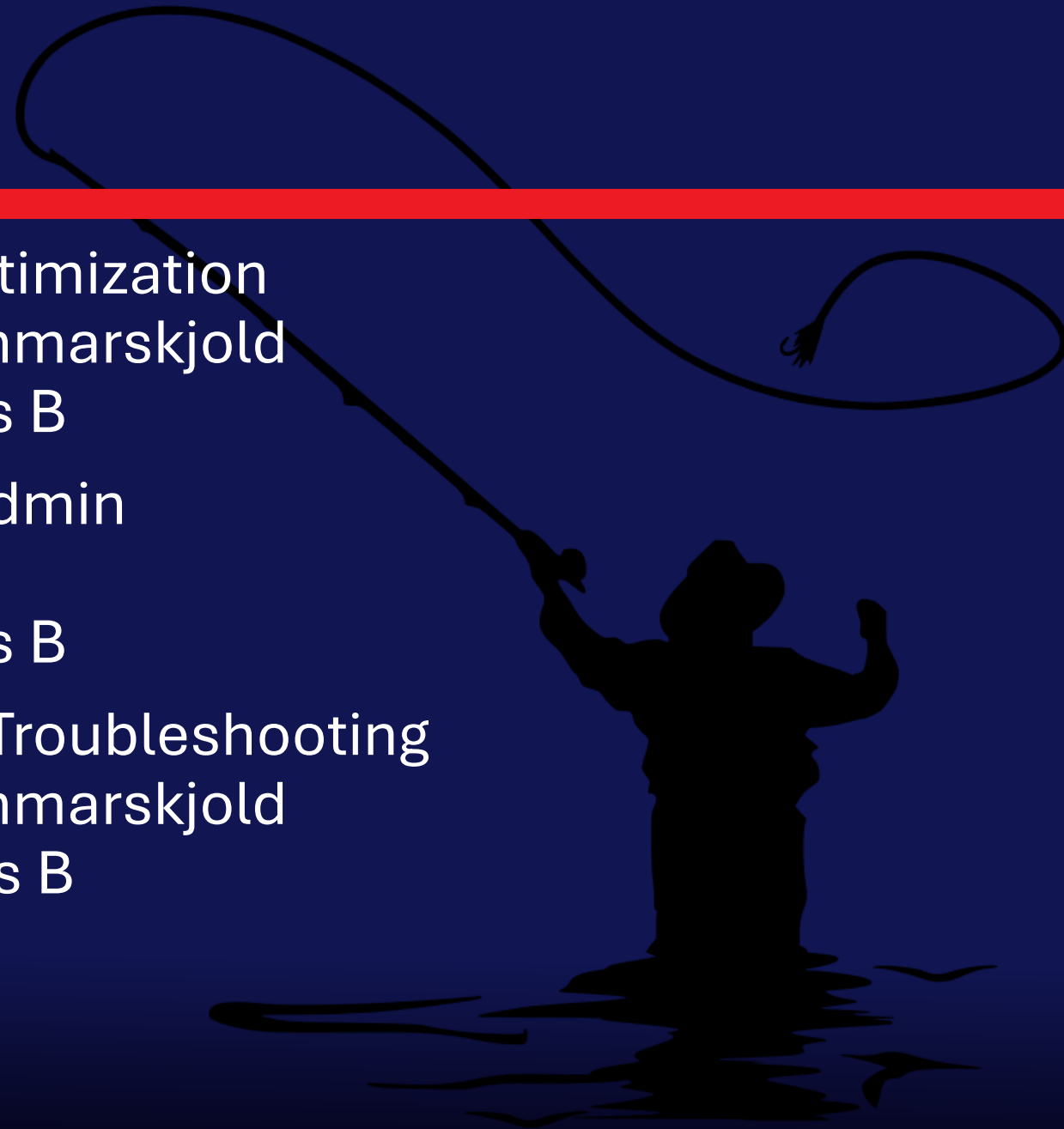
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Building your own lab



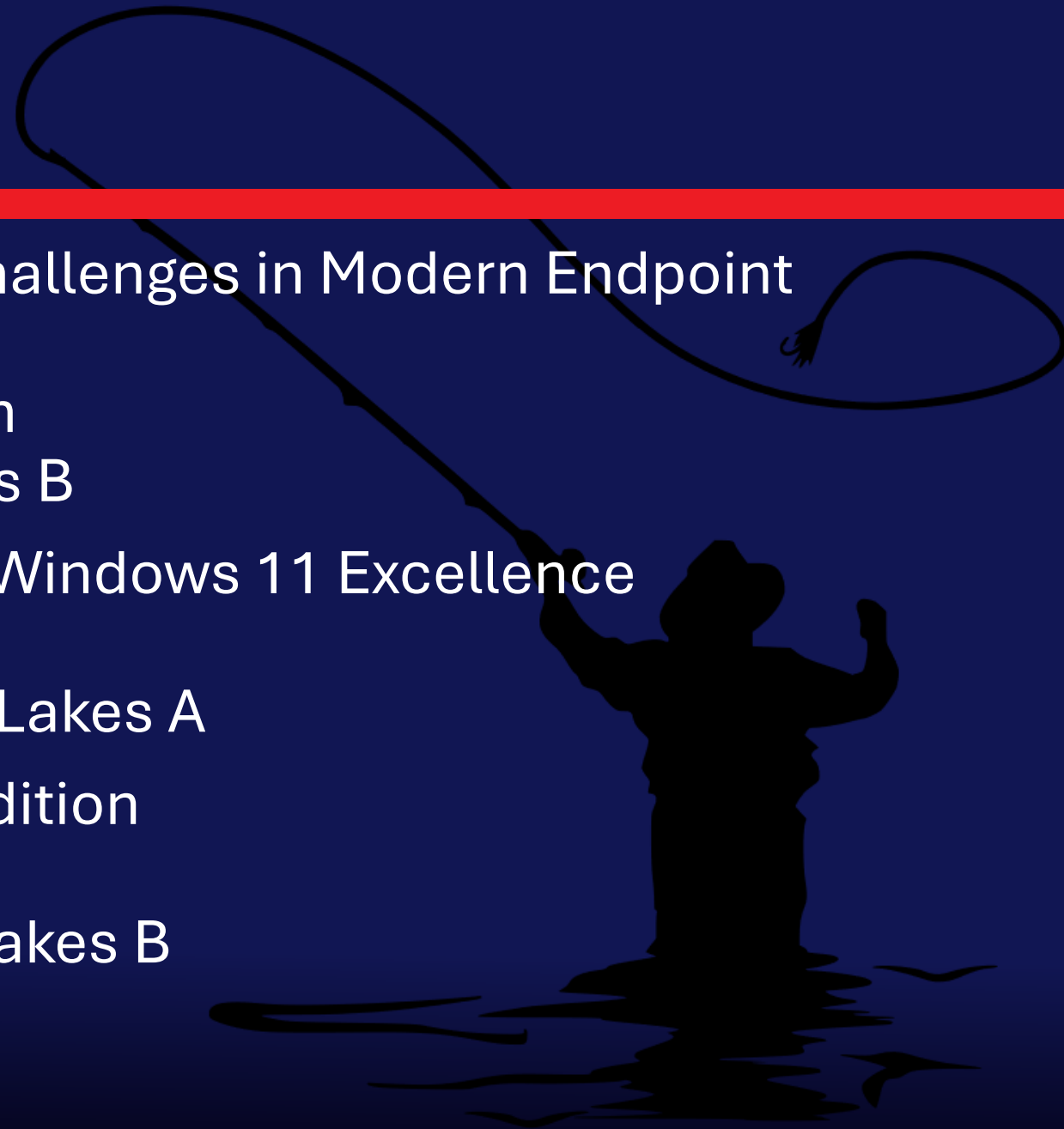
Other Sessions

- ♦ Maximizing Endpoint Delivery Optimization
Mattias Benninge & Andreas Hammarskjold
Monday, May 4th at 1:00 PM Lakes B
- ♦ OSD Essentials for the Modern Admin
Mike Terrill & Gary Blok
Monday, May 4th at 3:00 PM Lakes B
- ♦ Mastering Delivery Optimization Troubleshooting
Mattias Benninge & Andreas Hammarskjold
Tuesday, May 5th at 1:00 PM Lakes B



Other Sessions

- ◆ Ensuring Business Continuity: Challenges in Modern Endpoint Mgmt
Mike Terrill & Carl Johan Hederoth
Tuesday, May 5th at 3:00 PM Lakes B
- ◆ Beyond Deployment: Sustaining Windows 11 Excellence
Johan Arwidmark & Mike Terrill
Wednesday, May 6th at 10:00 AM Lakes A
- ◆ Secure Boot Certificates: 2026 Edition
Mike Terrill & Gary Blok
Wednesday, May 6th at 1:00 PM Lakes B

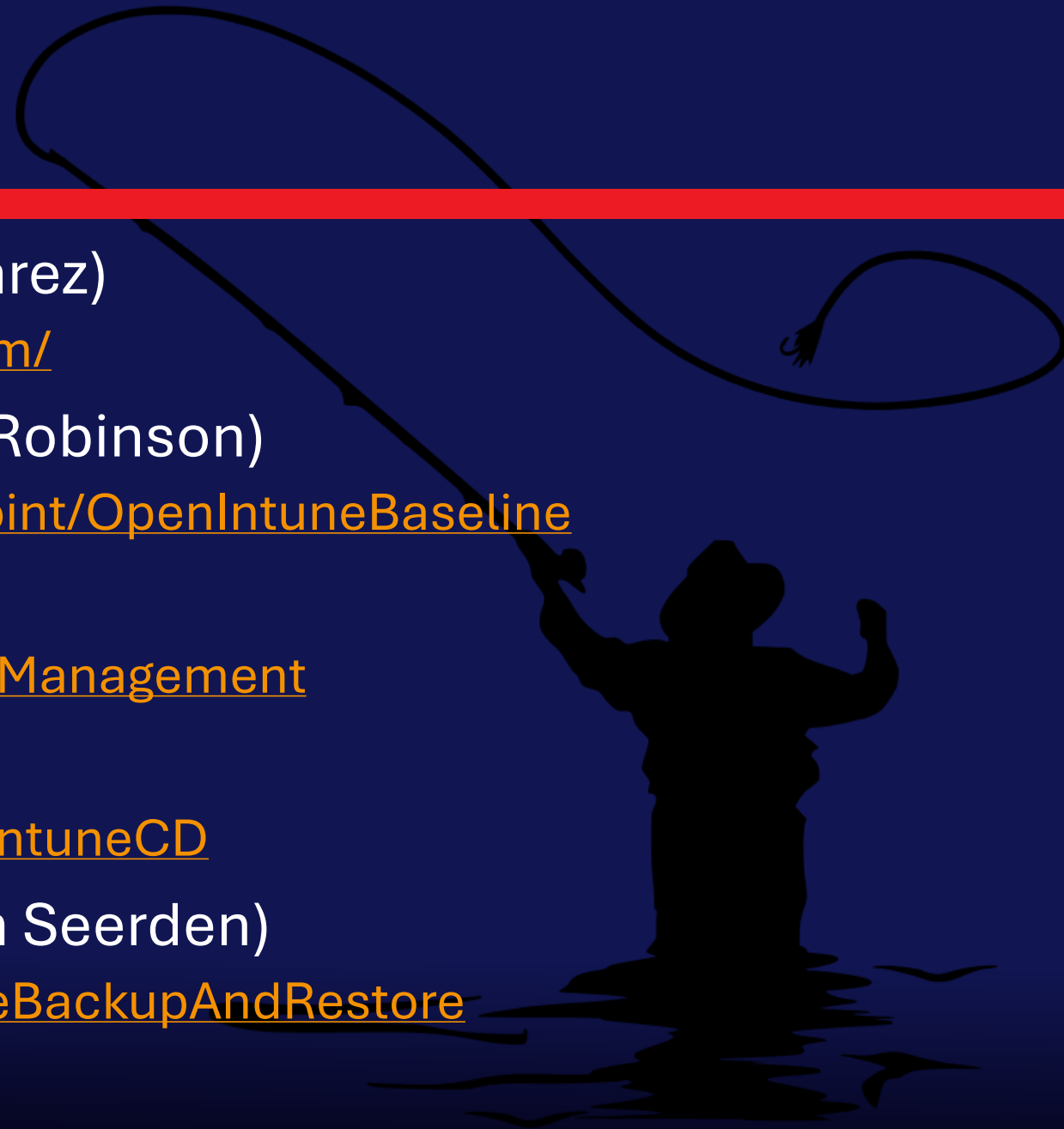


References

- ◆ Hydration Kit by Johan
 - ◆ <https://www.deploymentresearch.com/hydration-kit-for-windows-server-2022-sql-server-2019-and-configmgr-current-branch/>
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- ◆ [Creating Nested Hyper-V VMs - Deployment Research](#)
- ◆ [Using pfSense Community Edition as a virtual router for your lab environment - Deployment Research](#)

References

- ♦ Intune Hydration Kit (by Jorge Suarez)
 - ♦ <https://www.intunehydrationkit.com/>
- ♦ Open Intune Baseline (by James Robinson)
 - ♦ <https://github.com/skiptotheendpoint/OpenIntuneBaseline>
- ♦ Intune Management (by Micke K)
 - ♦ <https://github.com/Micke-K/IntuneManagement>
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References

- ◆ [miketerrill.net/Lab-Hyper-V at master · matterill/miketerrill.net · GitHub](https://github.com/matterill/miketerrill.net)
- ◆ [miketerrill.net/WinPE at master · matterill/miketerrill.net · GitHub](https://github.com/matterill/miketerrill.net)



SAVE THE DATES

Oct 25-28, 2026



May 2-6, 2027



Oct 10-13, 2027



Extended Q&A



Recast



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